

MCQs: MEDICAL GENETICS

CHAPTER 24: UNIVERSITY PRESS

ANSWER ALL QUESTIONS

1. What is the term used to refer to the full set of chromosomes from an individual?

- A. Genome
- B. Karyotype
- C. Chromosome set
- D. Gene map

2. Which syndromes are associated with an additional copy of chromosome 21?

- A. Edwards syndrome
- B. Patau syndrome
- C. Down syndrome
- D. Warkany syndrome 2

3. Which condition results from a trisomy of chromosome 18?

- A. Down syndrome
- B. Patau syndrome
- C. Edwards syndrome
- D. Turner syndrome

4. Which type of chromosomal abnormality involves gaining two extra chromosomes?

- A. Trisomy
- B. Tetrasomy
- C. Pentasomy
- D. Nullisomy

5. Which of the following describes Pentasomy?

- A. Loss of one chromosome
- B. Gain of one extra chromosome
- C. Gain of two extra chromosomes
- D. Gain of three extra chromosomes

6. What is the result of a complete loss of a pair of homologous chromosomes?

- A. Trisomy
- B. Tetrasomy
- C. Pentasomy
- D. Nullisomy

7. Which term refers to an organism with a single set of chromosomes?

- A. Diploid
- B. Polyploid
- C. Monoploid
- D. Euploid

8. Polyploidy refers to an organism with:

- A. A single set of chromosomes
- B. Two sets of chromosomes
- C. Three or more sets of chromosomes
- D. A missing set of chromosomes

9. Which chromosomal rearrangement involves the duplication of a chromosomal segment?

- A. Deletion

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- B. Duplication
- C. Inversion
- D. Translocation

- B. Inversion
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10. Which condition is caused by duplication of the gene encoding peripheral myelin protein 22 (PMP22) on chromosome 17?

- A. Down syndrome
- B. Charcot-Marie-Tooth disease type 1A
- C. Edwards syndrome
- D. Patau syndrome

14. Which chromosomal rearrangement involves a portion of one chromosome being deleted from its normal place and inserted into another chromosome?

- A. Inversion
- B. Duplication
- C. Translocation
- D. Insertion

11. What type of chromosomal rearrangement involves the exchange of segments between two different chromosomes?

- A. Inversion
- B. Duplication
- C. Reciprocal translocation
- D. Ring Formation

15. An isochromosome forms when:

- A. One arm of the chromosome is missing, and the remaining arm is duplicated
- B. Both arms of a chromosome are deleted
- C. A chromosome is split into two identical chromosomes
- D. A chromosome forms a ring structure

12. Which chromosomal abnormality only occurs with chromosomes 13, 14, 15, 21, and 22?

- A. Inversion
- B. Isochromosome
- C. Ring formation
- D. Robertsonian translocation

16. What type of chromosomal abnormality results in the formation of a hybrid gene product?

- A. Duplication
- B. Translocation
- C. Deletion
- D. Inversion

13. Which chromosomal abnormality occurs when a chromosome segment is broken, inverted, and reattached?

- A. Duplication

17. What is a diagram showing the ancestral relationships and transmission of

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genetic traits over several generations in a family called?

- A. Gene map
- B. Pedigree
- C. Family tree
- D. Karyotype

18. In a pedigree chart, what shape represents a female?

- A. Square
- B. Circle
- C. Diamond
- D. Triangle

19. What shape is used to represent a male in a pedigree chart?

- A. Circle
- B. Square
- C. Diamond
- D. Triangle

20. If the sex of an individual is unknown in a pedigree, what shape is used?

- A. Circle
- B. Square
- C. Diamond
- D. Triangle

21. In a pedigree, how are parents connected?

- A. By diagonal lines

B. By a horizontal line

C. By a vertical line

D. By a dotted line

22. What connects the parents to their offspring in a pedigree chart?

A. Horizontal line

B. Vertical line

C. Diagonal line

D. Curved line

23. How are siblings arranged in a pedigree?

A. From right to left according to birth order

B. From left to right according to birth order

C. From top to bottom according to birth order

D. In a circular pattern

24. How are twins represented in a pedigree chart?

A. By a vertical line connecting the parents

B. By horizontal lines connecting the siblings

C. By diagonal lines connecting the twins

D. By a circle and a square connected by a straight line

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25. Which term is used specifically for a male whose phenotype is of interest in a pedigree?

- A. Proposita
- B. Proband
- C. Propositus
- D. Offspring

26. Which term is used specifically for a female whose phenotype is of interest in a pedigree?

- A Proposita
- B. Propositus
- C. Proband
- D. Ancestor

27. In a pedigree chart, how is the proband indicated?

- A. By a circle
- B. By a square
- C. By an arrow pointing to the individual
- D. By a diamond

29. During which phase of mitosis does the nuclear envelope break down?

- A. Prophase
- B. Metaphase
- C. Anaphase
- D. Telophase

40. Cytokinesis is referred to as _____ -

- A. Division of the nucleus

B. Division of the chromosomes

C. Division of the cytoplasm

D. Division of the centromere

41. Which phase of interphase follows the S phase?

- A. G1
- B. G2
- C. G3
- D. M

42. What is the role of centrioles in mitosis?

- A. Replication of DNA
- B. Formation of the nuclear envelope
- C. Initiation and organization of the mitotic apparatus
- D. Cytokinesis

42. One feature of the telophase is _____

- A. Chromosomes align at the spindle equator
- B. Nuclear membranes reform around daughter nuclei
- C. Sister chromatids separate
- D. Chromosomes condense and become visible

43. Which cell cycle phase is characterized by cell growth and production of proteins and organelles?

- A. G1

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B. S

C. G2

D. M

44. During which phase do the chromosomes relax into their extended phase?

A. Prophase

B. Metaphase

C. Anaphase

D. Telophase

45. What term describes the two identical DNA molecules produced during the S phase?

A. Chromosomes

B. Sister chromatids

C. Centromeres

D. Nucleosomes

46. During the telophase, the _____ disappears

A. Nuclear membrane

B. Spindle fibers

C. Nucleolus

D. Centriole.

47. What is the primary function of the G2 phase?

A. DNA replication

B. Chromosome alignment

C. Preparation for mitosis

D. Cytokinesis

48. What phase immediately precedes anaphase in mitosis?

A. Prophase

B. Metaphase

C. Telophase

D. Cytokinesis

49. What is the main event of metaphase in mitosis?

A. Chromosome condensation

B. Chromosome alignment at the spindle equator

C. Separation of sister chromatids

D. Reformation of nuclear membranes

50. During which phase do progeny centrioles move apart?

A. Prophase

B. Metaphase

C. Anaphase

D. Telophase